MIL-R-20648C
22 September 1983
SUPERSEDING
MIL-R-20648B
26 August 1974

MILITARY SPECIFICATION

RACK, STORAGE, MOTION PICTURE, FILM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements of a storage rack for 16mm motion picture film on reels and in metal cans.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	- Preservation, Methods Of.
MIL-R-214	- Reel, Photographic Film, And Can Or Case, Photographic Film Storage (16mm and 35mm Film).
MIL-F-14072	- Finishes For Ground Electronic Equipment.
MIL-E-55585	- Electronics Equipment And Parts, Packaging Of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Electronics Research and Development Command, Combat Surveillance and Target Acquisition Laboratory, ATTN: DELCS-PE, Fort Monmouth, New Jersey 07703, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures And Tables For Inspection By Attributes.
MIL-STD-130	- Identification Marking Of US Military Property.
MIL-STD-454	- Standard General Requirements For Electronic Equipment.
MIL-STD-726	- Packaging Requirements Code.
MIL-STD-810	- Environmental Test Methods.

(Copies of specifications and Standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The rack shall be made of hot-rolled angle iron and sheet metal, 78 ±.06 inches (198.1 ±.15 cm) high, 48 ±.06 inches (121.9 ±.15 cm) wide and 16 ±.06 inches (40.6 ±.15 cm) deep. It shall consist essentially of two side panels, four longitudinal braces, shelves with supports, wire separators and legs, as specified herein.

3.2 First article. When specified, samples shall be subjected to first article inspection (see 4.3 and 6.2b).

3.3 Material. The material shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the rack to meet the requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.3.1 Recycled material. It is encouraged that recycled material be used, when practical, provided it meets the requirements of this specification.

3.4 Construction.

3.4.1 Side panel. The side panel shall consist of two angle iron uprights (corner posts), 1.5 x 1.5 x .125 inch (3.8 x 3.8 x .32cm) connected with a web of .03 inch (.08cm) thick sheet steel, plus 5 equally spaced angle iron shelf supports 1.0 x 1.0 x .125 inch (2.5 x 2.5 x .32cm). The shelf supports will be located as follows: one each across the top,

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center and bottom of the panel, with one equally spaced between top and center, and one between center and bottom. The entire side panel assembly shall be riveted together at those points where the shelf support angle and the vertical corner posts meet. Top and bottom of the sheet steel web shall be welded to the corner posts.

3.4.2 Longitudinal braces. Four braces shall be 1.0 x 1.0 x .125 inch (2.5 x 2.5 x .32cm) angle iron. Three braces, top, center and bottom shall form the back of the rack, by being fastened to the side panel uprights. One brace, at bottom, shall be similarly fastened across the front of the rack. A gusset type construction shall be used to fasten longitudinal braces to the uprights. Gussets shall be approximately .125 inch (.32cm) thick.

3.4.3 Shelves. Each of the 5 shelves shall consist of two 1.0 x 1.0 x .125 inch (2.5 x 2.5 x .32cm) angle iron shelf longitudinals. Each shelf longitudinal shall be 47.75 (+0 or -.06) inches (121.3 (+0 or -.15) cm) long with horizontal legs facing inward, centered and fastened to the shelf supports. Wire separators shall accommodate a minimum of 32 fully loaded reels in cans, in accordance with MIL-R-214, on each shelf.

3.4.3.1 Wire separators. Wire separators shall provide individual storage compartments for the reels and cans. Separators are to be of a modified inverted "U" shape, with vertical legs 6 (+0 or -.06) inches (15.2 (+0 or -.15) cm) apart to nest inside shelf support angles. The top of the "U" shaped separator shall extend 4 (+0 or -.06) inches (10.16 (+0 or -.15) cm) above the top edge of the shelf support angles. The separators shall be 7 gauge (.177 inch (.45cm) diameter) wire.

3.4.3.2 Identification strip. Each of the 5 shelves shall be equipped with an identification strip designed to hold individual index or identification cards for each reel of film. The rack shall be provided with a white (72 pound index or bristol) card for each location.

3.4.3.3 Plastic identification strip. Alternatively, when specified (see 6.2c), each of the 5 shelves shall be provided with a glossy, white platic vinyl or polystyrene strip having a minimum thickness of 0.01 inch (.025cm). The strips shall be fastened to the shelf longitudinals using a method such that the reels in cans can be identified by grease pencil markings on the strip. The strip shall remain permanently attached after subjection to all tests specified herein.

3.4.4 Rack legs. Cast iron or steel legs 2.0 x 2.0 x .25 inch (5.08 x 5.08 x .635cm) shall be fastened to the bottom of each upright.

3.4.5 Fastening. All parts shall be fastened together using welds, bolts, or rivets, or any combination thereof. All parts shall be fastened in accordance with the applicable paragraphs of MIL-STD-454.

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3.5 Environmental.

3.5.1 Humidity (moisture resistance). When tested as specified in 4.6.1.1, the rack shall exhibit no damage, such as corrosion, rust, blistering, swelling, or deterioration of parts and materials.

3.5.2 Vibration. When tested as specified in 4.6.1.2, the equipment shall show no splitting, separation, or bending of the elements of the rack including the fastenings thereof as a result of mechanical resonance.

3.5.3 Shock. When tested as specified in 4.6.1.3, the equipment shall show no loose separators, cracked or broken welds, loose rivets, misaligned or bent frames at the conclusion of the test.

3.6 Load capacity. When tested as specified in 4.6.2, the rack shall exhibit no broken parts, bent members, misalignment, broken, cracked, or loose fastenings at the conclusion of the test.

3.7 Separator. When tested as specified in 4.6.3, the separator shall not pull out of its pocket and neither the separators nor the formed retaining strips shall suffer any permanent deformation or other damage.

3.8 Finishes. Equipment shall be given protective finish in accordance with MIL-F-14072. This includes finishes of hardware, such as bolts, rivets, etc., and necessary touch-up after mounting. The final paint film on type 1 surfaces shall be final film E, semigloss, light gray enamel, in accordance with MIL-F-14072, table 1A (see 4.4).

3.9 Identification marking. All racks shall bear identification marking in accordance with MIL-STD-130.

3.10 Workmanship. The equipment shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

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a. First article inspection (see 4.3.2).

b. Inspection covered by subsidiary document (see 4.4).

c. Quality conformance inspections (see 4.5).

d. Packaging inspection (see 4.8).

4.3 First article.

4.3.1 First article units. When specified in the contract, the contractor shall furnish first article units of the complete Rack, Storage, Motion Picture, Film (see 6.3).

4.3.2 First article inspection. The first article inspection shall consist of the inspections specified in subsidiary documents covering the item listed in 4.4, and the inspections specified for group A and group C (see tables I and II respectively). The inspection shall be performed in the order as stated above. After completion of group C environmental tests, conforming units shall be reinspected and shall pass all group A inspection.

4.4 Inspection covered by subsidiary document. The following shall be inspected under the applicable subsidiary document as part of the inspection of equipment before packaging:

Item	Where required
Finish	3.8

4.5 Quality conformance inspection of equipment before packaging. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. Each unit shall be inspected for conformance to the inspections specified in table I. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels, and AQLs indicated in table I.

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TABLE I. Group A inspections.

Inspection	Requirement paragraph	Test method paragraph	AQL (percent defective)	
			Major	Minor
Visual & mechanical	3.4	4.7		
Workmanship	3.10	4.7	2.5	10

4.5.1.1 Order of inspection within group A. Group A inspection shall be performed in an order satisfactory to the Government.

4.5.2 Group B inspection. Not required.

4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed group A inspection. The inspection shall consist of the inspections specified in table II. Samples shall be selected in accordance with 4.5.3.1.

4.5.3.1 Group C sampling plan. One sample rack from every 25 units and fraction thereof produced shall be selected at random for all inspections in table II. The first sample selected shall be at the start of the contract from the first quality conformance inspection production lot (see 4.5.3).

4.5.3.2 Order of inspection within group C. Group C inspection shall be performed in the order shown in table II.

TABLE II. Group C inspection.

Inspection	Requirement paragraph	Test method paragraph
Humidity (moisture resistance)	3.5.1	4.6.1.1
Vibration	3.5.2	4.6.1.2
Shock	3.5.3	4.6.1.3
Load capacity	3.6	4.6.2
Separator	3.7	4.6.3

4.5.3.3 Group C failures. Actions required relative to group C failures shall be as specified (see 6.2).

4.5.4 Reinspection of conforming group C sample units. Unless otherwise specified, sample units which have been subjected to and passed group C inspection, may be accepted on the contract provided all damage is repaired and the same units are resubjected to and pass group A inspection.

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4.6 Tests.

4.6.1 Environmental.

4.6.1.1 Humidity (moisture resistance). The rack shall be tested in accordance with MIL-STD-810, Method 507.1, Procedure II. Inability of the rack to meet the requirements of 3.5.1 shall constitute failure of this test.

4.6.1.2 Vibration. The rack shall be tested in accordance with MIL-STD-810, Method 514.2, Procedure IX, Part 2. Inability of the rack to meet the requirements of 3.5.2 shall constitute failure of this test.

4.6.1.3 Shock. The rack in its assembled position shall be subjected to the following test: The rack shall be placed on a concrete floor and pushed over so that it falls flat on the floor. This test shall be repeated four times so that each side of the rack will make contact with the floor. Inability of the rack to meet the requirements of 3.5.3 shall constitute failure of the test.

4.6.2 Load capacity. The rack shall be set up for use on a concrete floor. Each shelf shall be loaded with 250 pounds (113.4kg) sand in bags, uniformly distributed over the length of the shelf. One end shall be raised 2 inches (5.08cm) and then released to drop on the concrete floor. The other end shall then be similarly raised and dropped. Inability of the rack to meet the requirements of 3.6 shall constitute failure of the test.

4.6.3 Separator. A pull of 8 pound-force (35.6N) shall be applied on the top of the separator for 2 minutes. After examining the separator for possible damage, a lateral or sidewise force of 8 pound-force (35.6N) shall be applied at the top of the separator for 2 minutes. Inability of the rack to meet the requirements of 3.7 shall constitute failure of the test.

4.7 Visual and mechanical inspection. Equipment shall be examined for conformance to the requirements of 3.3 through 3.10. Defects shall be classified as major or minor in accordance with MIL-STD-105.

4.8 Packaging inspection. Packaging inspection requirements specified herein are classified as follows:

a. First article inspection of packaging.

b. Quality conformance inspection of packaging.

4.8.1 First article inspection of packaging. Unless otherwise specified in the contract, first article inspection of packaging shall be in accordance with the unit pack design validation requirements of MIL-P-116.

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4.8.2 Quality conformance inspection of packaging.

4.8.2.1 Materials inspection. All materials to be used in packaging shall be inspected in accordance with the applicable material specification.

4.8.2.2 Preservation inspection. Inspection of preservation and interior markings shall be in accordance with group A and B quality conformance inspection requirements of MIL-P-116. Lot formation and sampling procedures shall be as specified therein.

4.8.2.3 Packing inspection. Inspection of packing and the marking for shipment and storage shall consist of the examinations specified in table III, "Packing inspection provisions." Lot formation shall consist of all packs made of the same materials during an identifiable period and submitted at one time for acceptance. Sampling procedures shall be in accordance with MIL-STD-105, using a single sampling plan and acceptable quality level of 4.0 percent defective.

TABLE III. Packing inspection provisions.

No.	Characteristic	Method of inspection
101	Intermediate container not as specified.	Visual
102	Improper closure of intermediate container.	Visual
103	Shipping containers not in accordance with specification.	Visual
104	Excessive cube.	Visual
105	Improper blocking and bracing.	Visual
106	Closure not in accordance with specification.	Visual
107	Weight and size exceed container limitations.	Weight & measure
108	Strapping not in accordance with specification, incorrectly applied, omitted.	Visual
109	Marking omitted, incorrect, or illegible.	Visual

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5. PACKAGING

5.1 Preservation. Preservation shall be as specified in MIL-STD-726, coded as follows:

5.1.1 Level A. 1Ø-1-1-ØØ-00-NS-X-ED-Ø-00-A.

5.1.2 Level B. 1Ø-1-1-ØØ-00-NS-X-ED-Ø-00-B.

5.2 Packing and marking. Packing and marking shall be in accordance with MIL-E-55585.

6. NOTES

6.1 Intended use. This equipment is intended for use in storage of fully loaded 16mm motion picture reels and cans.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. When a first article is required for inspection and approval (see 3.2, 4.3, and 6.3).

c. Whether plastic identification strips are required (see 3.4.3.3).

d. Group C failure action (see 4.5.3.3).

e. Level A or B preservation and packing (see section 5).

f. When first article inspection rough handling tests are not required.

g. When first article packaging inspection test reports require acquisition activity approval prior to production unit packing.

6.3 First article. When a first article inspection is required, the item will be tested and should be a first article sample. The first article should consist of the quantity of first article units specified in the contract. The contracting officer should include specific instructions in acquisition documents for examination, test, and approval of the first article.

6.4 Level B preservation. When level B preservation is specified, this level of protection will only be used under known favorable conditions during transportation, storage and handling.

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6.5 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ER
Air Force - 99

Review activity:
Air Force - 70
DLA - GS

Preparing activity:
Army - ER
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-20648C	RACK, STORAGE, MOTION PICTURE, FILM

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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